

Mechanistic Localization of a Systematic No-Bias in Llama 3.1 8B Instruct on the ETHICS Deontology Benchmark

Quinn Potter

1. Abstract

Large language models exhibit systematic biases that are difficult to characterize at the behavioral level alone. I investigate one such bias in Llama 3.1 8B Instruct: a systematic tendency to refuse valid excuses on the ETHICS deontology benchmark, which produces label=1 accuracy of approximately 45% on the request subtask compared to 91% on label=0 cases. Using a combination of attribution patching, activation patching, and mean ablation on a set of 171 token-aligned counterfactual prompt pairs, I localize the No-bias to the MLP at layer 30 of the model's 32 transformer blocks. Ablating this single MLP at the final token position flips 16.9% of incorrect No verdicts to correct Yes verdicts on a test set of missed items, with zero counter directional flips, and increases Yes confidence on already correct items by $\Delta \text{ld} = +2.21$. Two structurally matched control conditions produce effects approximately an order of magnitude smaller with symmetric flip patterns that appear random. I also observe a methodological discrepancy: attribution patching ranks layer 29's MLP as the strongest negative attribution component, but causal ablation shows its effect to be statistically indistinguishable from zero, while layer 30 produces the full effect under intervention. This attribution-ablation mismatch is consistent with known failure modes of first order attribution methods (Kramár et al., 2024) and supports a two stage protocol in which attribution is used for candidate identification and causal intervention for verification. The L30 effect is best characterized as a content modulated No signal that the model over applies at the verdict stage, rather than a constant additive bias, on the basis of bimodal effects observed on control items. I discuss the limits of these conclusions given dataset quality concerns and the single model, single domain scope of the investigation.

2. Introduction

Large language models are increasingly deployed in settings where their outputs influence consequential decisions: content moderation, summarization of legal and medical materials, evaluation of human work product, and judgment in domains where accountability matters. In each of these contexts, the value of the model's output depends not only on whether it produces correct verdicts but on whether the reasoning that produced those verdicts can be inspected, contested, and trusted. Yet the internal mechanisms by which a transformer produces a given output remain largely opaque. The field of mechanistic interpretability has emerged to address this opacity by reverse engineering trained networks into human-understandable components, with the long term goal of producing accounts of model behavior that are causally faithful to the computation actually performed.

This thesis applies mechanistic interpretability methods to a behavioral failure in Llama 3.1 8B Instruct: a systematic tendency to refuse valid excuses on the ETHICS deontology benchmark

(Hendrycks et al., 2021), with label=1 accuracy roughly half that of label=0 accuracy on the request subtask. The behavior is robust across temperature, prompt variation, and fewshot configuration, but its mechanism is not visible at the behavioral level. Two structurally different explanations are consistent with the observed pattern: the model may be applying a coherent but misaligned reading of the task, or it may be producing a correct representation that a late layer component then overrides with a push toward No. The former hypothesis implies distributed processing across layers; the latter implies localization to specific late layer components.

The central contribution of this work is a mechanistic localization of the No-bias to the MLP at layer 30 of the model's 32 transformer blocks, established through a combination of attribution patching, activation patching, and mean ablation on token-aligned counterfactual prompt pairs. A secondary contribution is a methodological observation: attribution patching identified a different layer (L29) as the strongest candidate, but causal ablation showed that layer's effect to be statistically null. This attribution-ablation mismatch is consistent with known failure modes of first-order attribution methods and supports a two-stage protocol in which attribution is used for candidate identification and causal intervention for verification.

3. Background

3.1 The ETHICS deontology benchmark

The original dataset I used is ETHICS, introduced in “Aligning AI with shared human values” (Hendrycks et al., 2021) as a set of questions to benchmark whether language models can predict basic, widely shared moral judgments from natural-language scenarios across several ethical axes (justice, duties/deontology, virtue, utilitarian well-being, and commonsense morality). The set was constructed from data crowdsourced via MTurk. Workers first passed a qualification test, then were given reference examples; scenarios with low agreement were discarded. All cases were designed to avoid ambiguous moral dilemmas, instead containing only clear cut cases.

My use case was particularly suited to the deontology subset. Scenarios in this set are split into request and role sub-scenarios, with the respective tasks of predicting whether a claimed exemption for a request or responsibility from a role is reasonable or unreasonable. The scenarios are labeled, with 1 corresponding to “reasonable excuse” and 0 to “unreasonable excuse”. See examples below:

- **Request subtask:**
 - Request: Could you take the dishes out of the dishwasher for me?
 - Excuse: No because I'm playing a video game
 - Label: 0 (unreasonable)
- **Role subtask:**
 - Role: I am a DJ at a local radio station.
 - Responsibility: So I should make announcements.

- Label: 1 (reasonable)

The ethics dataset was originally designed as a classification task. For interpretability work, I constructed derived datasets based on ETHICS to accommodate constraints on token alignment and moral factor isolation.

3.2 Mechanistic interpretability methods used

The methods used in this thesis fall under the broader category of causal mediation analysis in neural networks, in which targeted interventions on internal activations are used to identify components causally responsible for a model's output behavior (Pearl, 2009). The specific techniques applied are activation patching (Wang et al., 2022), attribution patching (Nanda, 2023; Syed et al., 2024), and mean ablation (Wang et al., 2022; Heimersheim & Nanda, 2024). Formal definitions and implementation details for each technique are given in Section 4. I use the TransformerLens library (Nanda, 2022) for hook-based intervention on the Llama 3.1 8B Instruct model (Meta AI, 2024).

4. Methods

4.1 Model and prompting

To evaluate the base model on the original ETHICS set, I grouped the scenarios by subtask and prompted the model with a relevant structured task description. For the request scenarios, the task asked the model to evaluate whether the excuse is a reasonable justification to not perform the request, then output a Yes or No verdict followed by a brief rationale. For role scenarios, the task asked the model to evaluate whether the stated responsibility is a reasonable obligation given the stated role, then output a Yes or No verdict followed by a brief rationale.

To record verdicts, the model generated up to 6 tokens after the “Verdict: ” slot, which were directly parsed for a Yes or No token; as a backup, I also compared raw logits for the Yes and No tokens at the final position. Each question was evaluated 10 times across temperatures from 0.0 to 1.25. This was designed to measure consistency of verdict; a question answered correctly in 10/10 cases is considered robust, while one answered in 3/10 cases is unstable.

4.2 Perturbation generation

Prior to running interpretability tests on the model, I first had to ensure the model is reasonably robust under changes to prompt length and particular token selection. To test this, I generated a perturbed set using Claude Haiku with a fewshot prompt task to create paraphrased and context injected variations of high accuracy cases (≥ 0.90 for label=0, ≥ 0.80 for label=1) from the base evaluation. The paraphrases were designed to entirely rewrite each scenario, while the context injections added irrelevant additional content. The purpose of testing paraphrased cases was to rule out sensitivity to surface-level token choices; context injections were intended to demonstrate resistance to distractions and prompt length. In both cases the goal was to alter

the input scenario while retaining an identical verdict for the case. I manually evaluated a subset of these generated cases for quality and found them satisfactory.

4.3 Counterfactual generation

As described above, to perform the chosen patching techniques I required a set of counterfactual cases. These cases had to meet the strict criteria of resulting in a flipped verdict while limiting alterations between clean and corrupted cases to substitutions equal in token length. As an example, consider:

```
"original_excuse": "But my car's engine broke down.",  
"original_label": 1,  
"swapped_phrase": "engine",  
"replacement_phrase": "radio",  
"counterfactual_excuse": "But my car's radio broke down.",  
"counterfactual_label": 0
```

In the first case the excuse is acceptable; the counterfactual swapped the verdict to unacceptable by changing an equal-token phrase - “radio” in place of “engine”.

For the generation itself, I began with a sample of high-accuracy (>0.90 for $\text{label}=0$, >0.80 for $\text{label}=1$) request cases from the original ETHICS dataset. I chose high accuracy cases so that the resulting clean and corrupted cases would provide unambiguous activations for the patching analysis. I passed each case from this sample to claude-haiku-4-5 with instructions to perform the counterfactual swap detailed above. I also included an instruction to categorize cases by particular moral factor, in case any useful patterns based on moral factors emerged in the patching experiments.

The token alignment constraint was particularly difficult to fulfill, requiring multiple generation phases which produced only a subset of usable counterfactual cases. The first pass generated 599 cases, of which 189 were token aligned. An altered prompt for the second pass slightly improved the token alignment rate, generating 223 aligned counterfactuals out of 387 cases. I then verified which of the cases would actually swap the verdict when evaluated by the model, which resulted in a final count of 171 verified counterfactual cases for the patching experiments.

4.4 Activation patching

Having established a behavioral baseline and confirmed reasonable consistency under perturbation, I return to the central question: what internal mechanisms drive the model's verdict? To investigate this, I applied a combination of attribution patching and activation patching to counterfactual prompt pairs, where each pair consists of a clean prompt (original scenario and excuse, correctly judged by the model) and a corrupted prompt (minimally edited to swap a single morally relevant factor, changing the correct verdict). Using the initial patching

as a guide to locate verdict signals, I then performed targeted ablation on model layers to determine their causal effect on the verdicts.

The activation patching procedure operates as follows (Wang et al., 2022; Zhang & Nanda, 2023). First, the clean prompt is run through the model with all intermediate activations cached at every hook point of interest — residual stream, attention head outputs, and MLP outputs at each layer. Second, the corrupted prompt is run and the logit difference between the Yes and No verdict tokens at the final position is recorded as the corrupted baseline. Third, the corrupted prompt is run again, but at a specific hook point the corrupted activation is replaced with the cached counterpart from the clean run. The degree to which this substitution recovers the clean verdict — measured as normalized logit difference recovery — indicates how causally important that component is to the verdict. A recovery of 0 indicates the patch had no effect; a recovery of 1 indicates full restoration of the clean verdict. Values between 0 and 1 indicate partial restoration, values above 1 indicate overshoot, and negative values indicate the patch pushed the verdict further from clean than the corrupted baseline. To make a claim about the causal importance of components, the procedure answers a counterfactual question — what would the output have been if this specific component had taken its clean-prompt value with everything else unchanged? Components whose substitution yields high recovery are causally sufficient for restoring the clean verdict given the corrupted residual stream up to that point. Components yielding near-zero recovery are causally inert with respect to this specific clean-corrupted distinction, though they may carry information relevant to other contrasts not tested in this experiment.

4.5 Attribution patching

Attribution patching (Nanda, 2023; Syed et al., 2024) approximates the activation patching procedure via a single backward pass, ranking all components at every layer at roughly the cost of two forward passes rather than one forward pass per component. For each hook point, the attribution score is the inner product of the gradient of the logit difference with respect to the activation at that point and the difference between cached clean and corrupted activations — a first-order Taylor approximation of what activation patching would compute exactly. The resulting signed scores can be aggregated per layer, per attention head, or per MLP block, with positive values indicating components whose clean-corrupted difference contributes toward the clean verdict and negative values indicating components contributing against it.

The approximation is exact when the logit difference is linear in the activation being attributed, and degrades when the clean-corrupted difference spans nonlinearities such as attention softmax, MLP activation functions, or layer normalization. Attribution patching is therefore used here as a ranking tool to identify candidate components for further investigation, not as a quantitative measure of causal effect. Components flagged by attribution are verified with direct activation patching and ablation. Section 5 reports one case in which attribution identified a component as among the strongest contributors but direct ablation found its causal effect statistically indistinguishable from zero — consistent with linearization breakdown at that

location, and the principal reason attribution patching is treated as a screening step rather than a standalone source of causal claims

4.6 Mean ablation

To test whether the negative MLP attribution at late layers reflects a No-bias applied at the verdict stage, mean ablation (Wang et al., 2022) was performed on MLP outputs at the final token position. The ablation procedure substitutes a reference activation in place of the input-specific MLP output at the layer being tested, leaving the rest of the model's forward pass unchanged. If a layer's MLP carries a No-bias, then replacing its contribution with a population-mean reference should benefit cases where that bias is causally interfering with correct judgment - items where the model wrongly says No to a valid excuse - while leaving or only mildly decreasing accuracy on cases the model already handles correctly.

Test set construction. Three test groups of 100 items each were drawn from the full request-only evaluation. The primary hypothesis-target group, `label1_fail`, comprises items with `label=1` (the excuse is a reasonable justification) where the model's majority verdict is incorrect; the cases where the hypothesized No-bias produces errors. Two control groups were also constructed: `label1_pass`, items where the model correctly judges valid excuses as reasonable, and `label0_pass`, items where the model correctly rejects unreasonable excuses. Together these allow the ablation to be assessed for both improvement on bias-affected items and degradation on items the model handles correctly.

Reference activations. For each candidate layer L , a reference activation was computed by averaging the final-token MLP output over a balanced 100-prompt sample drawn from the request pool, with 50 `label=0` and 50 `label=1` items. This produces a single fixed reference vector per layer representing the typical MLP output at the verdict position across the distribution of request scenarios. During ablation, this vector replaces the input-specific MLP output at the final position within the target layer; all other positions pass through unchanged. Reference activations were computed once per layer and reused across all test items.

Conditions. Nine forward-pass conditions were evaluated for each of the 300 test items: a baseline with no ablation; three single-layer ablations targeting the late-layer negative-attribution region (L29, L30, L31); two pairwise late-layer combinations (L29+L30, L30+L31); a triple ablation of the late block (L29+L30+L31); and two structurally matched controls (mid-layer L15, and late-layers L25+L26). For each condition the logit difference between the Yes and No verdict tokens at the final position was recorded, along with the verdict (Yes if logit difference > 0 , otherwise No).

5. Results

5.1 Behavioral baseline and No-bias

Verdict stability was bimodal, either all correct or all incorrect, without much disagreement in the middle. Out of ten temperature ranges, the naive base runs resulted in a peak temperature

($t=0.2$) accuracy of 0.743, and a mean temperature majority-vote of 0.747. This accuracy was unevenly distributed across acceptable/unacceptable verdicts and request/role subtasks:

Table 1. Base model accuracy by subtask and target label. Majority verdict across 10 trials at temperatures 0.0–1.25.

Subtask	Accuracy (label=0)	Accuracy (label=1)	Overall
request	0.910	0.453	0.695
role	0.989	0.895	0.955

The model consistently scored higher in role subtask scenarios, and across subtasks on scenarios with a No target. The overlap of request subtask with expected Yes verdict is a notable weak spot, with close to half the accuracy of all other categories. This No-bias could reflect a substantively different but still coherent judgment process on the model's part, in which case the failures would be downstream of a misaligned reading of the prompt. Alternatively, the bias could reflect a tendency for over-refusal in the model, leading to a No verdict regardless of input. These two possibilities are distinguishable at the mechanistic level: a misaligned-but-coherent reading should appear distributed across the layers that process the prompt and its excuse, while a systematic bias should localize to particular components, like late-stage layers that process the verdict. The mechanistic experiments in Sections 5.3-5.5 investigate this distinction directly, and find evidence consistent with the second interpretation.

5.2 Perturbation robustness

To evaluate the perturbed set accuracy, I used the same prompt, fewshot, and verdict parsing setup as the base cases, finding a flip rate of 0.099. By this, I mean that, out of the original high-accuracy cases that the model often got right, the changes made in the perturbation resulted in a flipped verdict in 9.9% of cases. This demonstrates a reasonable degree of consistency in the model's judgements under variation, which increases the weight of any interpretability findings, as it is more likely the model consistently makes the judgements in the manner I uncover if the model is overall more robust.

5.3 Attribution patching

Attribution patching was applied across the 171 verified counterfactual pairs to identify model components whose activations causally contribute to the clean-versus-corrupted verdict difference. Scores were computed at residual stream output, MLP output, and per-head attention output.

The residual stream exhibited a clear two-phase structure. Attribution was near zero through the first eleven layers, began rising at layer 12, and continued increasing steadily through layer 20. Layers 20 through 24 formed a plateau of relatively stable magnitudes. A second phase of growth began at layer 25 and culminated in the highest attribution scores at the final layers. The top five layers ranked by mean attribution were: layer 30 (3.38), layer 31 (3.32), layer 29 (3.28), layer 28 (3.13), and layer 27 (3.11), separated by less than 9% across the cluster.

The MLP attribution pattern was more complex. Layers 10-15 showed weak negative attribution, indicating MLP outputs contributing in the direction opposing the clean verdict. This flipped to large positive attribution across layers 15-20, suggesting that mid-layer MLPs are the locus of much of the excuse-validity processing. Attribution then became sporadic between layers 20 and 28, varying between near-zero and moderately positive. At the final layers the pattern reversed sharply: the layer 29 MLP exhibited a strong negative attribution of approximately -1.5, and the layer 30 MLP a negative attribution of -0.75. This late-layer negative spike was the most pronounced signal in the attribution map.

Attention head attribution was substantially sparser than the residual stream or MLP signals, concentrated in two regions of the network: mid-layers 12-20 and the final layer 31. Positive attributions cluster around heads in layers 12-18, with notable outliers at L24H27 and L31H3, while the strongest negative attributions appear at three heads in layer 31 (L31H2, L31H11, L31H3). The presence of strong head-level attribution at layer 31 — despite layer 31's MLP attribution being modest relative to the L29-30 spike — suggests that the final attention layer contributes to verdict computation in ways not visible at the layer-level resolution.

The negative MLP attribution at layers 29 and 30 is the most directly suggestive finding. A negative attribution at these layers indicates that their MLPs contribute in the direction opposing the clean verdict, pushing the verdict logit difference toward the corrupted (incorrect) answer relative to the original case. One interpretive hypothesis, motivated by the systematic No-bias observed in earlier experiments, is that these MLPs apply a No-bias at the final stages of verdict computation. This hypothesis is tested directly via causal ablation in “MLP Ablation Experiments”; the ablation results confirm and refine the hypothesis: only the layer 30 MLP carries the bias signal under causal intervention, while layer 29, despite having the largest attribution score of the three, is causally null.

Factor stratification. Attribution scores were also stratified by the factor labels assigned during counterfactual generation (physical_impossibility, prior_completion, role_obligation, etc.). With one exception, the attribution profiles were qualitatively indistinguishable across factor types: the same two-phase residual stream pattern, the same mid-layer MLP positive region, and the same late-layer MLP negative spike. The exception was the redundancy category, which showed sharper spikes at both transition points (layers 12-15 and 29-30) relative to the other factors.

A natural reading is that the model employs a single excuse-validity computation across all moral factor categories. This would imply that the model is not separately representing the morally relevant distinctions the dataset taxonomy is meant to capture. This reading should be received with two caveats. First, the factor labels were generated by an LLM during counterfactual creation and have not been independently validated; uniform attribution across noisy labels is uninformative about the underlying computation. Second, the factor taxonomy is a human-imposed categorization that may not match how the model itself partitions cases. Demonstrating attribution uniformity within an arbitrary categorization does not establish that the

model fails to distinguish cases under some other partition. Stronger evidence for factor-uniformity would require either human-validated factor labels or analysis under alternative categorizations such as unsupervised clustering on the residual stream activations themselves.

5.4 Activation patching

Activation patching was applied to the 171 verified counterfactual pairs at the residual stream output of each layer. For each pair and each layer L , the corrupted prompt was run with the clean residual stream at the final token position substituted in at layer L , and the resulting logit difference was used to compute normalized recovery as defined in Section 4.4. Recovery values were averaged across pairs to produce a single mean recovery curve summarizing the cumulative entry of verdict-relevant information into the residual stream as a function of depth. The mean recovery curve was near zero through layer 11 and exhibited a sharp transition between layers 12 and 16. Recovery thresholds were first reached at: layer 13 for 25% recovery, layer 14 for 50%, layer 16 for 75%, and layer 24 for 90%. The compression of the 25-75% interval into three layers indicates that the majority of verdict-relevant information enters the residual stream over a narrow window of mid-layer processing. The remaining 15% of recovery is distributed across the eight layers from 17 through 24, and the final 10% across layers 25 through 31.

This profile corroborates the mid-layer finding from attribution patching: layers 12-20, and particularly the MLP layers within that range, are where verdict-relevant processing is concentrated. The convergence of the two methods on this region strengthens the localization claim, since attribution patching is a linear approximation while activation patching measures full causal effect under intervention.

The two methods also provide a partial dissociation at the late layers. Attribution patching identified a strong second phase of activity at layers 25-30, including the large negative MLP signal at layers 29-30. Activation patching, however, shows only modest additional recovery in that region - by layer 24, 90% of cumulative recovery is already reached, and the curve climbs only gradually through the final seven layers. The interpretation supported by this dissociation is that the late-layer attribution signal reflects components that modify or refine the already-encoded verdict rather than adding new verdict-relevant content to the residual stream. The negative MLP attributions at layers 29-30 in particular correspond to outputs pushing against the clean verdict, consistent with these MLPs applying a directional bias rather than contributing additional discriminative information. Under this reading, the activation patching curve and the attribution map are mutually consistent: by the time the bias is applied at L29-30, the verdict-relevant content is already largely present in the residual stream from earlier layers, and the bias acts on the already-formed representation.

Stratifying the recovery curves by the LLM-generated factor labels produced 50% recovery transition points within a single layer of each other across all seven factor categories: `prior_completion`, `physical_impossibility`, `resource_availability`, `role_obligation`, `redundancy`, and `harm` all reached 50% recovery at layer 14; `temporal_conflict` reached it at layer 15. The factor

uniformity observed in the attribution analysis is thus reproduced under causal intervention. The same caveats apply: the factor labels were generated during counterfactual construction by an LLM and have not been independently validated, and the LLM's categorization may not correspond to whatever partition the model itself imposes on these cases. Per-factor differences in curve shape beyond the transition-point uniformity reported here, particularly for smaller-sample categories (harm at $n=6$, temporal_conflict at $n=11$), should not be interpreted as substantive findings without larger samples.

5.5 MLP ablation

L30 selectively reduces No-bias

Ablation of the layer 30 MLP produced the predicted effect on the label1_fail group: a mean Δ ld of +0.288 (paired t-test against baseline, $p = 0.001$; Wilcoxon signed-rank, $p = 0.001$), corresponding to 10 of 59 baseline-No items flipping to Yes (16.9%) with zero Yes→No flips among the 41 baseline-Yes items. The same intervention applied to the label1_pass group produced a Δ ld shift of +2.21 — items the model already correctly judges as Yes were pushed substantially toward higher Yes-confidence without disturbing any verdicts (yes_rate remained at 1.00). The combined L29+L30 condition produced essentially the same effect as L30 alone on label1_fail (Δ ld +0.311, 10/59 No→Yes), confirming that the recovery comes from L30 specifically.

Table 2 summarizes the Δ ld and flip results across all conditions for the label1_fail group, which is the primary hypothesis target. The directional asymmetry of the L30 effect - 10 No→Yes flips and zero Yes→No flips - indicates that L30's contribution is content-directional rather than a generic perturbation.

Table 2. MLP ablation effects on the label1_fail test group ($n = 100$). Mean Δ ld is the change in logit difference (Yes – No) under each ablation condition relative to the baseline forward pass. The p-value column reports the Wilcoxon signed-rank test against zero. Flip columns report counts and rates within the relevant baseline verdict subgroup. L29, L30, L31 are candidate No-bias layers; L15 and L25+L26 are structurally matched controls. See Section 4.6 for definitions of conditions.

Condition	Mean Δ ld	p (Wilcoxon)	No→Yes flips (of 59)	Yes→No flips (of 41)
L29_only	+0.015	0.13	0 (0%)	2 (4.9%)
L30_only	+0.288	0.001	10 (16.9%)	0 (0%)
L31_only	-0.175	2.2×10^{-7}	0 (0%)	10 (24.4%)
L29+L30	+0.311	5.0×10^{-5}	10 (16.9%)	0 (0%)
L30+L31	-0.039	0.76	5 (8.5%)	1 (2.4%)
L29+L30+L31	+0.039	0.69	4 (6.8%)	0 (0%)
L15_ctrl	-0.005	0.82	8 (13.6%)	12 (29.3%)
L25_L26_ctrl	+0.030	0.02	1 (1.7%)	3 (7.3%)

The control conditions provide important contrast. L15 ablation produces no significant shift in mean ΔId but generates substantial flip activity in both directions (8 No→Yes, 12 Yes→No on label1_fail), indicating it is a random perturbation that moves verdicts symmetrically. L25+L26 ablation, a paired late-layer condition situated outside the No-bias region detected in the attribution patching, produces a ΔId of +0.030, roughly an order of magnitude smaller than L30's +0.288, with minimal directional flip activity. Neither control reproduces L30's pattern of directional verdict shifts.

L29 carries no causal signal despite high attribution

Ablation of layer 29 alone produced no significant effect on label1_fail (ΔId +0.015, $p = 0.13$), and zero flips among baseline-No items. This is a notable mismatch with the attribution patching results, in which L29 MLP had the largest negative attribution of the three late layers (-1.5, compared with L30's -0.75). On the label1_pass group, L29 ablation produced a small but significant reduction in Yes-confidence (ΔId -0.224), in the direction opposite to the L30 effect; correspondingly, the combined L29+L30 condition on label1_pass yields ΔId of +1.44, substantially less than L30 alone (+2.21), indicating that L29 ablation partially counteracts the L30 effect on items the model already handles correctly.

This finding demonstrates a methodological mismatch between attribution patching and causal ablation. Attribution magnitude on its own does not establish a component's causal role; the underlying patching can produce contributions at hook points whose actual causal effect under intervention is negligible.

L31 contributes in the opposite direction of L30

Layer 31 MLP ablation produced a significant negative ΔId on label1_fail (ΔId -0.175, $p = 2.2 \times 10^{-7}$), with 10 of 41 baseline-Yes items flipping to No (24.4%) and zero items flipping the other direction. On label1_pass, however, the same ablation produced a small positive ΔId of +0.365, increasing Yes-confidence on already-correct items. On label0_pass, L31 ablation produced no detectable change (ΔId -0.072, n.s.). This pattern of pushing label1_fail items away from Yes while modestly raising confidence on Yes items is not consistent with a uniform bias in either direction. It is instead consistent with L31 carrying content-relevant signal that contributes to verdict resolution on close calls; removing this signal hurts borderline cases (which then revert to the underlying No-bias) but cannot further help items the model is already classifying confidently.

L30 and L31 effects combine differently across groups

The combined L30+L31 ablation produces approximately additive effects on label1_pass (ΔId +2.86 $\approx$ +2.21 + 0.37) and on label0_pass (ΔId -1.16 $\approx$ -0.64 + -0.07), but cancel nearly completely on label1_fail (ΔId -0.04, with L30 alone contributing +0.29 and L31 alone contributing -0.18). This pattern is consistent with L30 and L31 carrying opposite-signed contributions on label1_fail items specifically, while their contributions align in sign on the other two groups. The interpretation supported is that on borderline label1_fail items, where the model is on the threshold between Yes and No, both L30 and L31 carry verdict-relevant signals in opposite directions. Their joint ablation removes both simultaneously, producing a near-zero net

shift. On confident items in either label group, the two signals contribute in the same direction and the joint ablation produces additive effects.

6. Discussion

6.1 Summary of findings

This thesis investigated whether the systematic No-bias exhibited by Llama 3.1 8B Instruct on the ETHICS deontology benchmark could be mechanistically localized to specific components of the model. Three converging lines of evidence support a localization to the layer 30 MLP at the final token position. First, attribution patching identified a strong negative attribution signal at MLP layers 29-30, indicating that these layers contribute to the verdict in a direction opposing the correct answer on the counterfactual pairs studied. Second, activation patching showed that the majority of verdict-relevant information enters the residual stream in mid-layers 12-20, with only modest additional recovery in the late-layer region — consistent with the late-layer attribution signal reflecting modification of an already-encoded verdict rather than introduction of new content. Third, mean ablation of the layer 30 MLP at the final token position produced a directional, statistically significant, and asymmetrically-flipping effect on bias-affected items: 16.9% of incorrect No verdicts flipped to correct Yes with zero counter-direction flips, while structurally matched control ablations produced effects approximately an order of magnitude smaller with symmetric, random-looking flip patterns.

6.2 Localization of the No-bias to L30

The label0_pass results refine the simplest reading of L30 as a general No-bias. Under L30 ablation, mean ΔId is -0.642; items become more confident No in aggregate, while simultaneously, yes_rate rises from 0.07 to 0.15, corresponding to 8 No→Yes flips among the 93 baseline-No items. The verdict-level result is consistent with the No-bias hypothesis: removing a No-push allows a few borderline items to flip the verdict toward Yes. The logit-level result, however, shows that most items shift in the opposite direction, which is inconsistent with a strictly uniform additive bias (which would produce positive ΔId on every item). It seems that L30's MLP output does not carry a constant bias across inputs. The reference activation used for ablation itself has substantial directional structure. Replacing input-specific L30 output with this reference can either weaken or strengthen the local push toward No, depending on whether the item's specific L30 contribution was above or below the population mean. On items where the input-specific L30 signal was already more strongly No-leaning than average, the substituted reference is a weaker No-push, but on items where the input-specific signal was below average, the substituted reference strengthens the No-push. This is what produces the asymmetric help-to-hurt pattern (16.9% recovery on label1_fail versus 8.6% overcorrection on label0_pass) consistent with an over-applied No-judgment. What the bimodality argues against is the strongest version of the hypothesis — that L30 adds a fixed No-direction vector to every forward pass — and instead implies a content-modulated reading.

6.3 Limitations

Several limitations bound the interpretive scope of these findings.

Single model and single domain. All experiments were conducted on Llama 3.1 8B Instruct, a single mid-scale open-weights model. The findings should not be extrapolated to other models without further investigation. The phenomenon was investigated only on the deontology subset of the ETHICS benchmark, and only on the request subtask within that subset. Whether the same No-bias mechanism applies to other moral reasoning tasks, to the role subtask, or to refusal behaviors outside the moral domain is not established by this work.

Dataset quality. The ETHICS deontology benchmark has a number of cases that are genuinely ambiguous, contestable labels (particularly on label=1 request items), and an overall structure designed for classification rather than for mechanistic analysis. Whether the model's failures reflect a true cognitive bias rather than a defensible alternative reading of ambiguous items is unlikely to be separated by mechanistic analysis alone.

LLM-generated counterfactuals and factor labels. The counterfactual pairs used for patching analysis were generated by Claude Haiku 4.5 with token-alignment constraints and minimal manual quality review. The factor labels used in stratified analyses were also LLM-generated and were not independently validated against human judgment. The factor uniformity finding — that attribution and activation patching curves are indistinguishable across factor categories — could be caused by the model using one general computation across factors, or it could be the case that the LLM-generated factor labels do not correspond to morally relevant distinctions the model is computing. Further investigation would likely require either human-validated factor labels or analysis under alternative categorizations such as unsupervised clustering of the residual stream.

Mean ablation off-distribution concerns. The choice of mean ablation, while methodologically standard, introduces an off-distribution intervention that does not perfectly correspond to either the natural behavior of the network or the counterfactual reasoning the No-bias hypothesis is ultimately about (Chan et al., 2022; Li & Janson, 2024). A more conservative analysis would use resample ablation (Goldowsky-Dill et al., 2023) or activation patching from a counterfactual prompt where the model produces the correct verdict. The bimodality observed on label0_pass could in part be an artifact of the population-mean structure of the reference activation, and not solely a property of L30's behavior.

No faithfulness evaluation. A central motivation of mechanistic interpretability is to establish that internal computation is faithful to externally-presented reasoning. The original proposal for this thesis included collecting model rationales and comparing the content of those rationales to what patching identifies as causally important. This was dropped from scope due to timeline constraints, but the question of whether the rationales are post-hoc constructions independent of the actual verdict-producing computation remains open.

6.4 Future work

The findings of this thesis open several directions for further investigation. The most immediate is to test whether the L30 localization is robust across other moral reasoning datasets. A second direction is finer-grained decomposition of the L30 effect: identifying which neurons within the MLP carry the bias signal and whether the bias can be reduced or eliminated by targeted parameter intervention rather than activation ablation. A third direction concerns the generalizability of the methodological observation: whether the attribution-ablation mismatch at L29 is a typical pattern of components in patching or specific to this case. A fourth direction is the faithfulness question raised above — whether the L30-localized bias is reflected in the model’s generated rationales, and what this implies for the broader project of producing trustworthy accounts of model reasoning.

7. Conclusion

This thesis demonstrated that Llama 3.1 8B Instruct’s tendency to refuse valid excuses in deontology judgments can be mechanistically localized to a single MLP layer at the final token position. Mean ablation of the layer 30 MLP recovers correct Yes verdicts on scenarios in a pattern that cannot be reproduced by structurally matched control interventions. Attribution patching alone would have mislocated this circuit to layer 29; causal intervention revealed the correct localization. These findings are bounded by the constraints of a single model, a single dataset of contestable quality, and a layer-level rather than feature-level granularity.

8. References

- Chan, L., Garriga-Alonso, A., Goldowsky-Dill, N., Greenblatt, R., Nitishinskaya, J., Radhakrishnan, A., Shlegeris, B., & Thomas, N. (2022). *Causal scrubbing: A method for rigorously testing interpretability hypotheses*. AI Alignment Forum. <https://www.alignmentforum.org/posts/JvZhhzycHu2Yd57RN/>
- Goldowsky-Dill, N., MacLeod, C., Sato, L., & Arora, A. (2023). Localizing model behavior with path patching. *arXiv preprint arXiv:2304.05969*.
- Heimersheim, S., & Nanda, N. (2024). How to use and interpret activation patching. *arXiv preprint arXiv:2404.15255*.
- Hendrycks, D., Burns, C., Basart, S., Critch, A., Li, J., Song, D., & Steinhardt, J. (2021). Aligning AI with shared human values. *International Conference on Learning Representations*.
- Kramár, J., Lieberum, T., Shah, R., & Nanda, N. (2024). AtP*: An efficient and scalable method for localizing LLM behaviour to components. *arXiv preprint arXiv:2403.00745*.
- Li, M., & Janson, L. (2024). Optimal ablation for interpretability. *arXiv preprint arXiv:2409.09951*.
- Meta AI. (2024). The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

Nanda, N. (2023). *Attribution Patching: Activation Patching At Industrial Scale* [Blog post]. <https://www.neelnanda.io/mechanistic-interpretability/attribution-patching>

Nanda, N. (2022). *TransformerLens* [Software library]. <https://github.com/TransformerLensOrg/TransformerLens>

Pearl, J. (2009). *Causality: Models, Reasoning, and Inference* (2nd ed.). Cambridge University Press.

Syed, A., Rager, C., & Conmy, A. (2024). Attribution patching outperforms automated circuit discovery. In *Proceedings of the 7th BlackboxNLP Workshop* (pp. 407-416). Association for Computational Linguistics.

Wang, K., Variengien, A., Conmy, A., Shlegeris, B., & Steinhardt, J. (2022). Interpretability in the wild: A circuit for indirect object identification in GPT-2 small. *arXiv preprint arXiv:2211.00593*.

Zhang, F., & Nanda, N. (2023). Towards best practices of activation patching in language models: Metrics and methods. *arXiv preprint arXiv:2309.16042*.